LLM: Alignment & RLHF & PPO

Modelni alignment lashtirishni asosiy 2 hil usuli bor:

1. DPO



2. Traditional RLHF



RLHF: model chiqargan 2 ta javobdan, user yaxshirogini tanlaydi. Natijada modelni yaxshilash mumkin

SFT da instruction-response juftliklar bulsa, RLHF da human preference signallari buladi. va qanday javob berish emas (bu SFT ni vazifasi) balki qaysi javob yaxshiroq ekanligini urgatadi.

Nega RLHF ishlatamiz deganda, userlarga qulayroq javob kerak. SFT ni vazifasi faqat tugri javob berish holos u bilmaydiki qulaylik jihatidan qaysi javob "tugri" ligini. Shu sababli RLHF ni ishlatamiz.

Nega RLHF mukammal emas: user tomonidan notogri tanlov bulsa modelga katta tasiri buladi va natijada modelham notogri urganadi.

Reward hacking: model yaxshi ball olish uchun notugri yul tutishi. Misol model foydali javob berish urniga, uzun javob yozib yuqori ball olishga harakat qiladi

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PPO: murakkabroq uqitish va reward orqali modelni boshqarish kerak bulganda ishlatiladi

SFT modelni 2 ta copy qiladi:

- reference model (muzlangan/frozen): rivojlantirilmaydi shunchaki ishlatilinadi
- policy model (PPO bilan update): yakuniy final model buladi

Full RLHF loop:

Prompt olinadi → Policy javob chiqaradi → Reward model baholaydi (SFT or API) → Reference bilan taqqoslanadi → Value (pos/neg) hisoblaymiz → PPO update

Feature	Traditional RLHF (PPO)	DPO
Base model	SFT model	SFT model
Preference data	Yes	Yes
Alohida reward model	Yes	No
RL loop (PPO)	Yes	No
Memory models	4 ta	2 ta
Optimization	Reward-based policy optimization	Preference-based objective
Murakkablik	High	Lower